

Power Series Solutions to Nonlinear Ordinary Differential Equations

MATH 689 Homework 1

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1. Adapted from problem B.3a on p. 204:

$$f' = \frac{-2x}{(1+x^2)^2}f, \quad f(0) = e, \quad x \in [0, 5] \quad (1)$$

1a. We modify the code in Example1.m to solve the problem numerically using RK4 via the following code in MATLAB

MATLAB

```
dx=0.0001; %STEP SIZE
x=0:dx:5; %DOMAIN OF INDEPENDENT VARIABLE
f=x; %PREALLOCATE MEMORY FOR DEPENDENT VARIABLE (will be overwritten below)
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=exp(1); %this seems silly now but is a placeholder for higher-order ODEs
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)) ;
    k3=F(u+k2*dx/2,x(n)) ; k4=F(u+k3*dx,x(n)) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color="y");
hold on
%IF OBTAINABLE, PLOT ANALYTICAL SOLUTION
fA=exp(1./(1 + x.^2)); % use .* and .^ and ./
plot(x,fA,'LineWidth',2,'DisplayName','Analytical Solution')
legend show; xlabel('$x$', 'Interpreter','latex','FontSize',20)
ylabel('$f$', 'Interpreter','latex','FontSize',20)
% 1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x); f=u;
dudx=[-2*x*u/((1+x^2)^2)];
end
```

1b. Classify the ODE (4 classifiers if nonlinear, 6 classifiers if linear) The ODE is has dependent variable f , independent variable x , first order, linear, homogeneous, variable coefficient, non-autonomous.

1c/d. Using separation of variables we obtain the analytical solution as

$$f = \exp\left(\frac{1}{1+x^2}\right) \quad (2)$$

Proof. Write f' as $\frac{du}{dx}$ and it follows that using separation of variables we have

$$\begin{aligned}\frac{du}{dx} &= \frac{-2xu}{(1+x^2)^2}, \quad t = 1+x^2 \\ \int \frac{du}{u} \frac{1}{t} dx &= \int \frac{-dt}{t^2} \\ \ln(u) &= \frac{1}{t} + C \\ f &= \exp\left(\frac{1}{1+x^2} + C\right) \\ f &= C_0 \exp\left(\frac{1}{1+x^2}\right) \quad \text{with } C_0 = 1\end{aligned}\tag{3}$$

□

1d. Numerical and Analytical Solution Plot

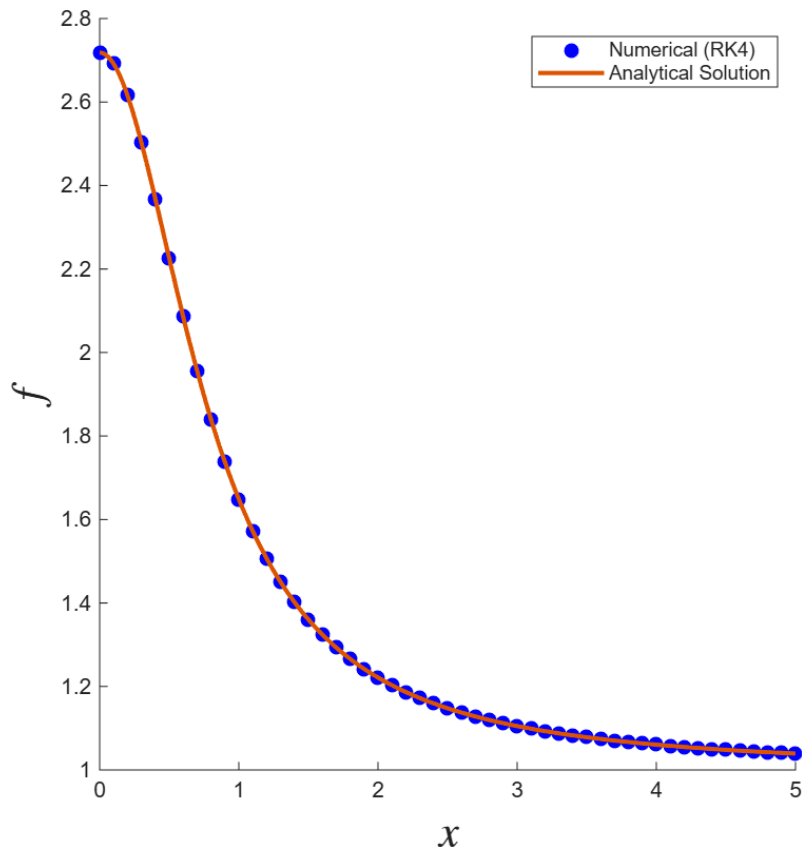


Figure 1: solution to (1)

2. Adapted from problem B.3c on p.204:

$$f' + f = e^{x^3}, \quad f(0) = 1, \quad x \in [0, 1] \quad (4)$$

2a. We modify the code in Example1.m to solve the problem numerically using RK4 via the following code in MATLAB

MATLAB

```
dx=0.00001; %STEP SIZE
x=0:dx:1; %DOMAIN OF INDEPENDENT VARIABLE
f=x; %PREALLOCATE MEMORY FOR DEPENDENT VARIABLE (will be overwritten below)
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=1; %this seems silly now but is a placeholder for higher-order ODEs
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)) ;
    k3=F(u+k2*dx/2,x(n)) ; k4=F(u+k3*dx,x(n)) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color="g");
hold on
%IF OBTAINABLE, PLOT ANALYTICAL SOLUTION
% fA=exp(1./(1 + x.^2)); % use .* and .^ and ./
% plot(x,fA,'LineWidth',2,'DisplayName','Analytical Solution')
legend show; xlabel('$x$', 'Interpreter','latex','FontSize',20)
ylabel('$f$', 'Interpreter','latex','FontSize',20)
%1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x); f=u;
dudx=[exp(x.^3)-u];
end
```

2b. Classify the ODE.

The ODE has dependent variable f , independent variable x , and is first order, linear, non-homogeneous, constant coefficient, non-autonomous.

2c. Using the integrating factor we are able to solve f in terms of integrals. Namely, if we have

$$f' + m(x)f = t(x) \quad (5)$$

then

$$f = \frac{1}{\exp[\int m(x)dx]} \left(\int t(x) \exp \left[\int m(x)dx \right] dx + K \right) \quad (6)$$

2d.

Proof. Using (6) we have

$$f = \frac{1}{e^x} \left(\int e^{x+x^3} dx + K \right) \quad (7)$$

then with $f(0) = 1$, we get $K = 1$, thus

$$f = \frac{1}{e^x} \left(\int e^{x+x^3} dx + 1 \right) \quad (8)$$

□

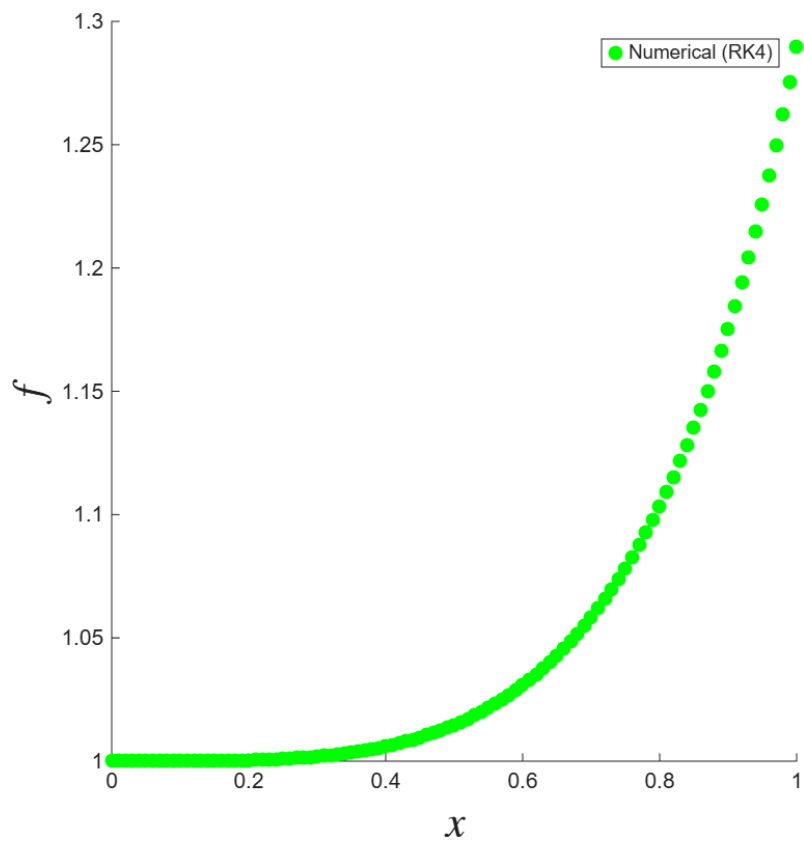


Figure 2: Solution to (4)

3. Adapted from problem B.3e on p. 205:

$$f'' + f = 0, \quad f(0) = 0, \quad f'(0) = 1, \quad x \in [0, 10] \quad (9)$$

3a. We modify the code in Example2.m to solve the problem numerically using RK4 via the following code in MATLAB

MATLAB

```
clear;clc;clf
%%Class Example 2 on 1/18/2024
dx=0.0001; %STEP SIZE
x=0:dx:10; %DOMAIN OF INDEPENDENT VARIABLE
f=x; %PREALLOCATE MEMORY FOR DEPENDENT VARIABLE (will be overwritten below)
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=[0 1];
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)+dx/2) ;
    k3=F(u+k2*dx/2,x(n)+dx/2) ; k4=F(u+k3*dx,x(n)+dx) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)', Color = "r");
hold on
%IF OBTAINABLE, PLOT ANALYTICAL SOLUTION
fA=sin(x); % use .* and .^ and ./
plot(x,fA,'LineWidth',2,'DisplayName','Analytical Solution')
legend show; xlabel('$x$', 'Interpreter','latex','FontSize',20)
ylabel('$f$', 'Interpreter','latex','FontSize',20)
% 1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x)
f=u(1); v=u(2);
dudx=[v -f];
end
```

3b. Classify the ODE. The ODE has dependent variable f , independent variable x , second order, linear, homogeneous, constant coefficient, and autonomous.

3c. Recall that if $f, f', f^{(2)}, \dots, f^{(n-1)}$ are all continuous functions and $f^{(n)}$ is a piecewise continuous function then,

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - \sum_{k=0}^{n-1} s^k f^{(n-1-k)}(0) \quad (10)$$

Notice that (9) is in terms in the form of $\sum_{1 \leq i \leq n} f^{(i)} = G(x)$ then we can deduce that

$$\begin{aligned} \mathcal{L}\left\{\sum_{1 \leq i \leq n} f^{(i)}\right\} &= \mathcal{L}\{G(x)\} \\ \sum_{1 \leq i \leq n} \left[s^i F(s) - \sum_{k=0}^{i-1} s^k f^{(i-1-k)}(0) \right] &= \int_0^\infty e^{-st} G(x) dt \end{aligned} \quad (11)$$

We can factor $F(s)$ and obtain our solution from the inverse Laplace transform of $F(s)$

$$\begin{aligned}
F(s) \sum_{1 \leq i \leq n} s^i &= \int_0^\infty e^{-st} G(t) dt + \sum_{1 \leq i \leq n} \sum_{0 \leq k \leq i-1} s^k f^{(i-1-k)}(0) \\
F(s) &= \frac{1}{\sum_{1 \leq i \leq n} s^i} \left[\int_0^\infty e^{-st} G(t) dt + \sum_{1 \leq i \leq n} \sum_{0 \leq k \leq i-1} s^k f^{(i-1-k)}(0) \right] \\
f(x) &= \mathcal{L}^{-1} \left\{ \frac{1}{\sum_{1 \leq i \leq n} s^i} \left[\int_0^\infty e^{-st} G(t) dt + \sum_{1 \leq i \leq n} \sum_{0 \leq k \leq i-1} s^k f^{(i-1-k)}(0) \right] \right\}
\end{aligned} \tag{12}$$

3d. Using our derivation from (12) we have

$$\begin{aligned}
&\mathcal{L}\{f'' + f = 0\} \\
&s^2 F(s) - sf(0) - f'(0) + sF(s) - f(0) = 0 \\
&s^2 F(s) - 1 + sF(s) = 0 \\
&(s^2 + 1)F(s) = 1 \\
&F(s) = \frac{1}{s^2 + 1} \\
&f(x) = \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 1} \right\} \\
&f(x) = \sin(x)
\end{aligned} \tag{13}$$

We plot the the Numerical solution using (RK4)

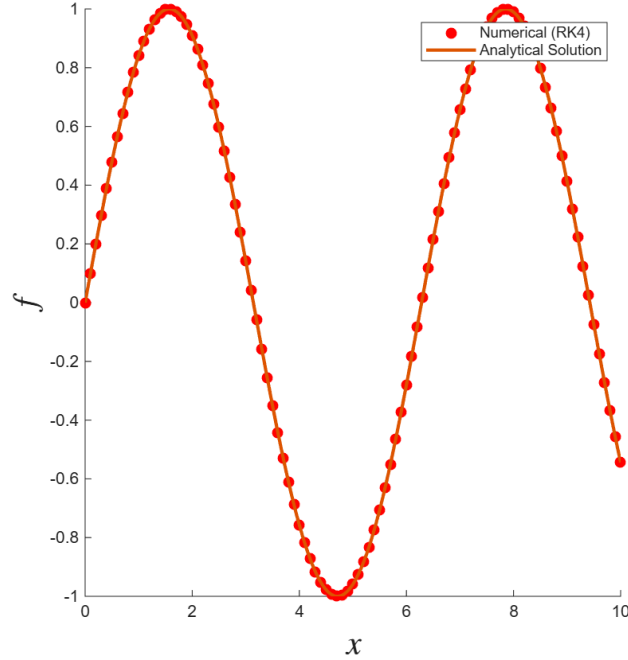


Figure 3: Solution to (9)

4. The Flierl-Petviashvili model [1] (used to in the description of Jupiter's red spot) has no known analytical solution in terms of elementary or (named) special functions.

$$f'' + \frac{f'}{x} - f - f^2 = 0, \quad f(0) = -2.392 \dots, \quad f'(0) = 0, \quad x \in [0, 10] \quad (14)$$

4a. We modify the code in Example2.m to solve the problem numerically over the domain in MATLAB

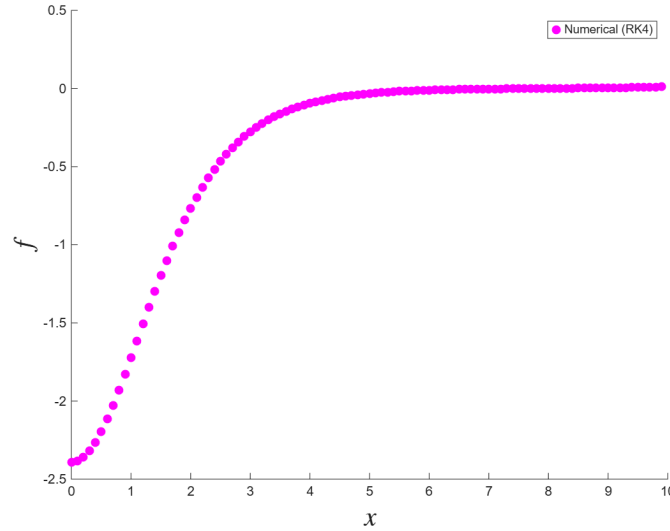


Figure 4 : Solution to (14)

MATLAB

```
dx=0.0001;
x=dx:dx:10;
f=x;
u0=[-2.392, 0];
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)+dx/2) ;
    k3=F(u+k2*dx/2,x(n)+dx/2) ; k4=F(u+k3*dx,x(n)+dx) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color= "m");
hold on
legend show; xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)
function dudx = F(u,x)
f=u(1); v=u(2);
dudx=[v f^2 + f - v/x];
end
```

4b. Classify the ODE.

The ODE in (14) has independent variable x , dependent variable f and is order 2, nonlinear, and non-autonomous.

5. MATH 689 Problem: A version of the Blasius boundary layer problem (problem B.3g on p. 305) is given below. This problem also has no known analytical solution in terms of elementary or (named) special functions.

$$f''' + ff'' = 0, \quad f(0) = 0, \quad f'(0) = 0, \quad f''(0) = 1, \quad x \in [0, 8] \quad (15)$$

5a. We modify the code in Example2.m to solve the problem numerically over the domain in MATLAB

We transform the third order ODE into a system of 3 equations.

$$\frac{d}{dx} \begin{pmatrix} f \\ v \\ w \end{pmatrix} = \begin{pmatrix} v \\ w \\ -fw \end{pmatrix}, \quad \begin{pmatrix} f(0) \\ v(0) \\ w(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (16)$$

MATLAB

```
dx=0.0001; %STEP SIZE
x=0:dx:8; %DOMAIN OF INDEPENDENT VARIABLE
f=x; %PREALLOCATE MEMORY FOR DEPENDENT VARIABLE (will be overwritten below)
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=[0 0 1];
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)+dx/2) ;
    k3=F(u+k2*dx/2,x(n)+dx/2) ; k4=F(u+k3*dx,x(n)+dx) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color= "m");
hold on
function dudx = F(u,x)
f = u(1); v = u(2); w = u(3);
dudx = [v w -f*w];
end
```

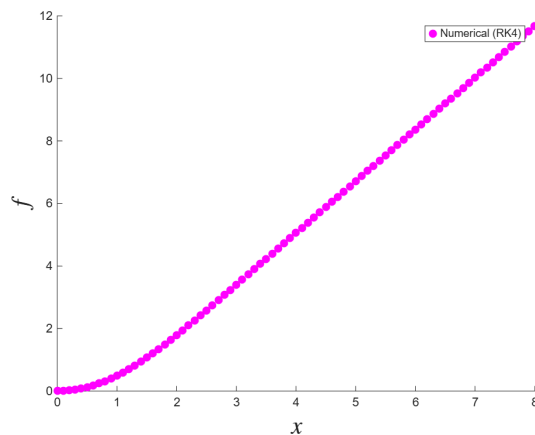


Figure 5: Solution to (15)

5b. Classify the ODE. The ODE in (15) has dependent variable f , independent variable x , and is 3rd order, nonlinear, and autonomous.

Reference

- [1] G. R. Flierl , *Baroclinic solitary waves with radial symmetry*, Dyn. Atmos. Oceans, 3 (1979), pp. 15-38.